

The Formal Register

Every symbol declared once, every claim numbered, every dependency drawn: the corpus stated so that it can be checked rather than read.

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What this volume is for. The Monograph tells the framework as one continuous argument and is meant to be read. The Observational Standard holds it to the measured record. Systems of Interpretation records where it grips and where it slides off. This volume does none of those things. It states the framework as a formal object: one notation, numbered axioms, numbered definitions, numbered results with their hypotheses and their proofs or citations, a certified numerical register, and an explicit dependency graph. A reader who wants to check whether the chain is sound should be able to do it here without reading the story at all.

What this volume is not. It is not a new derivation and it introduces no result that is not already carried by the Monograph or by a paper in the corpus. Where a proof is long it is stated compactly and the full derivation is cited by paper number. Where a claim is conditional the condition is written into the statement rather than into a footnote. Nothing here is graded higher than the corpus grades it elsewhere, and if this volume and any other document disagree, the Freeze Manifest of the current printing governs.

PART ONE

Notation

Every symbol used anywhere in the corpus is declared here, once. Symbols not on this list do not appear in the framework's own claims.

ϕ	the self-description fixed point, $\phi^2 = \phi + 1$
r	the retained fraction of one act of self-description, $1/(2\phi)$
u	the outward fraction, $1 - r$
$W_L W_B W_M$	the three weights: direct, related, held
τ	an ordered operational traversal; also proper time where marked
$[\tau]$	the equivalence class of traversals defining a recoverable object
Λ	the carrier lattice; Λ^* its dual
$\Phi(E_8)$	the 240 roots of E_8 ; $\alpha \beta \gamma$ individual roots
C	the system-preserving cluster projection (conditional expectation)
K	the decay generator; $B = \ker K$ the readable algebra
$P_X P_T P_I$	the geometry projectors: space, local clock, closure
V	the five-component cluster field
e, g	coframe and metric

D, F	covariant traversal operator and its curvature
G	the clustered gravity operator
M	the mass operator
Xi	the global carrier closure residue
l	the carrier-to-physical length conversion
hbar	the quantum of phase

PART TWO

Axioms and postulates

These are the commitments. Everything numbered in Part Four descends from them and from nothing else, which Part Six verifies.

- A1** Systemhood. A system is a system: it may hold differences internally without ceasing to be one thing. AXIOM.
- A2** The ground. The first distinction does not land in a waiting space; the holding of the distinction is what a where is. POSTULATE, stated in Movement Two rather than hidden.
- A3** The two-sided reading. A self-description reads its mixed relation in both directions, outward against held and held against outward. POSTULATE. This single commitment fixes the factor of two in r , forces the carrier's self-duality, and forces its evenness.
- A4** Saturation. An accumulating single channel reaches a condition where one channel can no longer carry what it has taken on. POSTULATE. Where that condition sits is OPEN.
- A5** Conservation of the readout. Nothing exposed at a closure event vanishes from the total of continuing face, outgoing record, and the ordering relation between them. POSTULATE.
- A6** Admissible reduction. A system-preserving reduction is unital, idempotent, completely positive, bimodular over the readable algebra, and covariant under equivalent carrier descriptions. These five conditions are the definition, not a choice among options: any map failing one of them deletes rather than decays.

PART THREE

Definitions

- D1** Distinction. That which prevents complete sameness. Primitive.
- D2** Identity. The recoverability of a distinction through admissible change.
- D3** Relation. The operation through which distinct contents affect, compare or transform one another without being erased.
- D4** The seven doings. Distinguish, hold, close, answer, change, order, record. The constitutive grammar.
- D5** Traversal stages. Persist, distinguish, relate, propagate, resolve, return, close. The grammar of one act passing through a world of finished things. Related to D4 by the noun rule: the first stage is the first three doings completed.
- D6** Object. A stable equivalence class of closed traversals. Objecthood is retrospective.
- D7** The update. The two-channel step carrying (resolved, held) to (resolved plus held, resolved).
- D8** Channels. The outward share u and the retained share r , with u plus r equal to one.

- D9** The three weights. The terms of $(u + r)$ squared: W_L equals u squared, W_B equals $2ur$, W_M equals r squared.
- D10** Crossing. One act of self-description enacted across a channel. Its price is the retained fraction r ; see T6.
- D11** Record. A distinction that remains discriminable and recoverable under the relevant future dynamics. Two-valued: there is no partially recoverable record.
- D12** Observer. Any system in which a distinction of another system becomes physically related and persistently retained. Mechanical observer: an observer with staged architecture, being a prepared ready state, a channel-defining coupling, propagation, resolution, amplification, and record production.
- D13** Carrier. The structure on which a self-describing record is written, required to equal its own dual.
- D14** Primitive crossing. The smallest nonzero admissible carrier displacement. In E8 these are exactly the roots, of squared norm two.
- D15** Cluster projection. The reduction C from carrier observables onto the readable subalgebra, satisfying A6.
- D16** Decay generator. K , the difference between doing nothing and applying C . Minimal form I minus C . See T14 for why the minimal form is not always the one meant.
- D17** Readable algebra. B , the part of the carrier algebra the decay generator leaves alone. This is the physical theory.
- D18** Geometry projectors. Mutually orthogonal P_X , P_T , P_I onto external distinction, local causal order and system closure, summing to the geometry block. The physical projector is P_X plus P_T ; P_I is retained as closure data and excluded from traversable directions.

PART FOUR

Results

Each result carries its hypotheses, its grade, and either a compact proof or a citation to the paper that carries the full derivation. A result stated as **CONDITIONAL** names its condition in the statement itself.

- T1** The seven doings admit a unique linear order. **PROVED**. Nothing is held before it is drawn, nothing closes before it is held, nothing answers before there is a closed thing to answer, nothing changes except by answering, nothing orders except as change follows change, nothing is recorded before it has happened. The dependency relation is a strict partial order with a unique linear extension. Monograph, Movement Two.
- T2** The minimal update is unique. **PROVED**. Requiring an update that adds and never subtracts, carries exactly one clean copy of each channel forward, and promotes the present resolved amount to the next reserve, admits exactly one operator. Movement Four.
- T3** The stable ratio is the golden fixed point. **PROVED**. Requiring the update to preserve the ratio between channels while changing overall size yields ϕ^2 equals ϕ plus one, which has exactly one positive solution. Movement Four.
- T4** The retained fraction is $1/(2\phi)$. **PROVED** given A3. The two-sided reading measures the retained side with a factor of two; u and r are then pinned with no adjustable quantity. Movement Four.
- T5** The three weights exhaust the whole. **PROVED**. Two channels squared give four combinations; the two mixed terms are the same pair read in opposite order and count once when reading direction is irrelevant. The three terms sum to one. Movement Five.

- T6** The quantised crossing. PROVED given D₁₁. A crossing either closes, at price exactly r , or fails to traverse. Intermediate prices would require a partial closure, and D₁₁ admits none. Exactness follows from the discreteness of closure rather than from tuning. Paper 18.
- T7** The traversal price in closed form. PROVED. r cubed equals one over eight phi cubed equals root five minus two, all over eight. Movement Seven; the carrier derivation of the exponent is T₁₀.
- T8** The carrier is E8. CONDITIONAL on A₃. Positive-definite even unimodular lattices exist only in ranks divisible by eight; at rank eight there is exactly one, namely E8, with 240 roots. At rank sixteen there are two and at rank twenty-four there are twenty-four, so the uniqueness is a property of the first realised rank. Movement Nine; lattice classification borrowed with its name attached.
- T9** Evenness descends from the two-sided reading. PROVED given A₃. A relation read outward and back counts its self-pairing twice, so carrier norms fall on the even integers. Positive-definiteness follows separately from D₁, since an element pairing to zero with itself is a distinction making no difference to itself. This reduces T8 from three stipulations to one. Movement Nine, new in the Fourth Printing.
- T10** The minimal closed carrier history is triangular. PROVED. For roots α and β with inner product minus one, $\alpha + \beta$ is a root; setting γ to its negative gives a sum of zero. Immediate reversal is cancellation and encloses nothing, so three is minimal. With T6 the weight of that history is r cubed. Paper 20.
- T11** The root cell area is root three. PROVED. The parallelogram spanned by two roots at that angle has area the square root of four minus one; the closed triangle is half of it. This identifies the root three carried since Movement Nine as a covolume. Paper 20.
- T12** The golden ring pairing. PROVED numerically to machine precision. Projected onto the Coxeter plane the 240 roots fall into eight rings of thirty whose radii pair off, four pairs, each in the ratio phi, with residual below one part in ten to the fifteenth. Research Archive; reproducible from the Cartan matrix and the eigenvectors of the Coxeter element.
- T13** The physical algebra is the kernel of the decay. PROVED given A₆. System identity cannot decay, so it lies in the kernel; what the generator suppresses becomes unreadable; the readable algebra is exactly the fixed algebra of the reduction. Paper 20; Movement Ten.
- T14** Minimal and general decay generators are distinct. PROVED. The minimal generator $I - C$ is idempotent, hence has spectrum in the two-point set, and its adjoint action has eigenvalues in minus one, zero and one. That suffices for T₁₃ and for the closure theorem, and it does not suffice for a graded descent or for a mass spectrum, which would collapse to two values. The physical generator is any positive operator annihilating the identity whose kernel is B . Paper 20, Appendix; new in the Fourth Printing.
- T15** The local clock mode is single. PROVED. The record accumulates along one monotone order; two independent orders of accumulation would be two records and therefore two systems. Movement Eleven, from the arrow of Movement Thirteen.
- T16** The closure mode is single. PROVED. There is one enacted closure whose record is inhabited; the closure mode counts enacted closures. Movement Eleven, from the one-world result of Movement Twenty-Three.
- T17** The inherited form is Lorentzian. CONDITIONAL on the opposite closure roles of spatial opening and causal completion. The inherited quadratic form is P_X minus P_T , up to overall sign and the conversion factor, which is the unique minimal nondegenerate form admitting null directions. Movement Twelve.
- T18** Second witness for T₁₇. PROVED, cited. Any bijection preserving the causal order of a flat continuum of more than two dimensions is a Lorentz transformation composed with a translation and a dilation, the dilation being the unit convention. Alexandrov and Zeeman; Paper 19. Independent of T₁₇'s route and reaching the same signature.

- T19** Superposition is forced. PROVED. If two actions are admissible and no physical event has distinguished one, selecting either adds information the system does not contain. The description must retain both. Movement Seventeen.
- T20** Complex phase is forced. PROVED. Additive action under sequential composition, unit modulus under reversible carriage, and continuity admit exactly one family of weights, the rotations, giving the exponential of the action in units of the quantum of phase. Movement Seventeen.
- T21** The Born pricing is forced. PROVED, cited. Requiring the same physical effect to carry the same probability in every complete measurement context admits exactly one pricing, the squared magnitude. Gleason, with the extensions of Busch and of Caves, Fuchs, Manne and Renes for the low-dimensional cases. Movement Eighteen.
- T22** Closed carriage is unitary. PROVED, cited. A transformation preserving every transition probability is unitary or antiunitary; continuity with the identity excludes the second. Its generator exists and is self-adjoint. Wigner; Stone. Movement Eighteen.
- T23** The boundary weight is the interference term. PROVED as arithmetic. Expanding the squared magnitude of a sum of two unresolved alternatives at primitive magnitudes u and r gives 0.477457514, 0.427050983 and 0.095491503, summing to one: exactly W_L , W_B and W_M . The cross-term, belonging to neither alternative alone, is interference. CONDITIONAL as physics on the primitive magnitudes being u and r . Movement Eighteen, new in the Fourth Printing.
- T24** The three weights structure the gravitational account. PROVED as algebra. The square of a defect formed from curvature weighted by u against realised area weighted by r has three terms and no others: the closed curvature record, the curvature-realisation cross-term, and the realised volume. Local gravitational dynamics reside in the cross-term. PROPOSED as physics for the identification of that cross-term with ordinary gravitational dynamics. Movement Nineteen; Paper 20.
- T25** Sector classification by one operator. PROVED given T14. Carrier operations commuting with the decay preserve the clustered identity and are gauge; operations connecting distinct decay sectors are matter when they close into a cycle reproducing their readable identity; failure to commute measures the cost of persistence and is mass. Modes commuting with the decay are massless. Movement Twenty; Paper 20.
- T26** Retention over replacement. PROVED given D7. A persistent self-description must join each new resolution to the record rather than overwrite it, or its later states are not records of the same system. Movement Twenty-Six.
- T27** Records are single, and which record is unreadable. PROVED. A state in which both alternatives remain coherently present is not discriminable as either and is therefore not a record; singleness is the definition of one. Any internal answer to which outcome would itself be a record produced by a resolving event, so the question regresses and the selection is not readable from within the record domain. Paper 18.
- T28** No recollapse while the record stands. CONDITIONAL on A5. Contraction of the causal domain toward coincidence would press resolved distinctions back into unread simultaneity, which is overwrite by geometry and is forbidden by T26. Paper 18.

PART FIVE

The numerical register

Every constant the framework uses, in closed form and to twelve significant figures. No value on this list is fitted, and no value is obtained by comparison with a measurement. All are reproducible from the Research Archive.

ϕ	$(1 + \sqrt{5})/2$	1.61803398875
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r	$1/(2 \phi) = (\sqrt{5}-1)/4$	0.309016994375
u	$1 - r$	0.690983005625
W_L	u^2	0.477457514063
W_B	$2ur$	0.427050983125
W_M	r^2	0.0954915028125
	$W_L + W_B + W_M$	1.000000000000
r^3	$(\sqrt{5} - 2)/8$	0.0295084971875
$1 - r^3$	retained, one crossing	0.970491502813
$(1-r^3)^3$	retained, three crossings	0.914061068091
$\sqrt{3}$	root cell parallelogram	1.73205080757
$\sqrt{3} / 2$	closed triangle cell	0.866025403784
r^{240}	global carrier residue	3.94246077122e-123
lambda	$(4 \pi / \sqrt{3}) r^{240}$	2.86033313614e-122

Two of these deserve a note. The traversal price is written as a surd throughout the corpus because a decimal in that position reads as something fitted, and the closed form shows what it is: three copies of one retained fraction, multiplied because the channels are crossed in series. The standing vacuum readout is a computed value with a stated construction and no adjustable quantity anywhere in its map; numerical resemblance to a measured value is not promoted to derivation anywhere in this corpus.

PART SIX

The dependency graph

The Monograph's governing promise is that nothing is slipped in by hand. That promise is a claim about structure and can therefore be checked rather than believed. Draw every result of Part Four as a node, draw an edge from each result to every result, axiom or definition it uses, and ask whether the graph contains a cycle. A cycle would mean a result standing on itself somewhere along the chain, which is the formal shape of the failure the promise forbids.

31 nodes, 33 edges, no cycles, five ungrounded roots

The graph is acyclic. Its ungrounded roots, the nodes standing on nothing but themselves, are exactly five: the two-sided reading A_3 , the definition of a record D_{11} , the definition of the cluster projection D_{15} , the unique order of the seven doings T_1 , and the minimal update T_2 . Everything else in the framework descends from those five by the edges drawn. The two results at the base of the graph, T_1 and T_2 , are themselves proved from the stated dependency requirements rather than assumed.

This is the strongest structural statement the corpus can make about itself, and it is worth being exact about what it does and does not establish. It establishes that no claim in the framework is used in its own derivation. It does not establish that the claims are true, and it does not establish that the roots are the right roots. It rules out one specific failure, circularity, completely and checkably. The remaining ways to be wrong are all in Part Seven.

PART SEVEN

Proof obligations

What the framework owes, stated formally, with the reason each item is outstanding. The corpus does not object to open questions; it objects to questions left open without anyone having looked. Each item below has been looked at.

- O1** A concrete representation of the cluster projection C on the E_8 carrier. OPEN because a representation is a computational input rather than an argument. Most of the remaining obligations wait on this one.

- O2** That rank P_X equals three. OPEN, conditional on O_1 , because it must emerge from the commutant structure of the decay generator in that representation. Constraint: the physical projector must have rank four and the surviving local symmetry must be the one T_{17} derives.
- O3** The low-energy gauge algebra, graded matter modules, chirality, families and anomaly freedom. OPEN, conditional on O_1 . These are representation-theoretic outputs and are the intended content of the calculation programme.
- O4** That a complete carrier closure passes each of the 240 roots exactly once. OPEN. The global vacuum residue rests on it. This obligation is finite and combinatorial, neither empirical nor representational, and is therefore the most attackable item on this list.
- O5** The integral joining the growth of the record to the measured expansion history. OPEN. The zeroth-order clause is proved; what remains is a calculation under the standing no-free-parameter rule.
- O6** The audited closure counts of the two measurement pipelines that disagree about the expansion rate. OPEN and empirical. The framework's declared count is three, its prediction is exposed rather than supported, and the current standing is a pull of about half a standard deviation.
- O7** A running, self-holding system returning quantised retention at powers of r under a convention declared before the count. OPEN and empirical. This is the framework's own stated condition of defeat and cannot be settled at a desk.

One representation, one number that follows from it, one catalogue that follows from both, one combinatorial lemma, and three measurements. That is the complete list. Everything else in the chain from A_1 to T_{28} is proved, conditional on a condition written into its own statement, or dissolved as a unit convention.

PART EIGHT

Where each result lives

The Monograph carries the argument as a story and grades every claim in place. The Observational Standard carries what may honestly be said about a measured system. Systems of Interpretation carries the record of the framework being thrown at real systems, including its misses, which fall in the regimes its own grammar forbids. The Papers as Issued carry the full derivations: the semantic layer and the logic of physical action in Papers 13 to 17, the exactness of the traversal price in Paper 18, the closing ledger in Paper 19, and the forward closure from carrier to Spectra boundary in Paper 20. The Research Archive carries the data, scripts and figures, and every number in Part Five reproduces from it.

This volume carries none of that and duplicates none of it. It carries the shape.

PART NINE

Concordance of movement numbers

The Fourth Printing inserted four movements into the Monograph, so every movement from the tenth onward carries a number one to four higher than it did in the Third Printing. Papers 18 and 19 were issued against the earlier numbering and are preserved as issued rather than silently edited, in keeping with the corpus discipline that corrections belong in new printings rather than in altered files. A reader following a cross-reference from either of those papers should read it through this table. Paper 20, issued with this printing, uses the current numbering throughout.

Third	Fourth	Movement	
-	10	The descent	(new)
-	11	What survives	(new)
-	12	The shape of the world	(new)
10	13	The record of the whole	
11	14	The equation as a finished journey	
12	15	The action and its double	
13	16	The container	
14	17	The unresolved world	
15	18	The bookkeeping of chance	
16	19	The carrying of sameness	
-	20	The one operator	(new)
17	21	The extended traversal	
18	22	The watcher's machinery	
19	23	The ocean	
20	24	The architecture of one world	
21	25	The luminous wound	
22	26	The wound that could not forget	
23	27	What it claims, and what it leaves open	

Movements One to Nine are unchanged in number. Movement Nine is rewritten in content, and Movements One to Eight carry over verbatim apart from three sanctioned edits recorded in the Freeze Manifest.